

Software Engineering

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Introduction to Spring Framework

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Roadmap

- Introduction to Spring
- Dependency Injection and IoC
- Bean
- AoP
- Module Architecture



What Is Spring Framework?

- Spring is the most popular application development framework for Java enterprise
- Open source Java platform since 2003.
- Spring supports all main application servers and JEE standards
- Spring handles the infrastructure so you can focus on your application
- Current version: 5.0.X



What does Spring offer?

■ Dependency Injection

- Also known as IoC (Inversion of Control)

■ Aspect Oriented Programming

- Runtime injection-based

■ Portable Service Abstractions

- The rest of spring
 - ORM, DAO, Web MVC, Web, etc.
 - Allows access to these without knowing how they actually work



Dependency Injection

- The technology that actually defines Spring (Heart of Spring).
- Dependency Injection helps us to keep our classes as independent as possible.
 - Increase reuse by applying low coupling
 - Easy testing
 - More understandable

An injection is the passing of a dependency (a service) to a dependent object (a client). Passing the service to the client, rather than allowing a client to build or find the service, is the fundamental requirement of the pattern.



Dependency Injection and Inversion of Control (IoC)

In software engineering, inversion of control (IoC) describes a design in which custom-written portions of a computer program receive the flow of control from a generic, reusable library.

- The Inversion of Control (IoC) is a general concept, and it can be expressed in many different ways and dependency Injection is merely one concrete example of Inversion of Control.



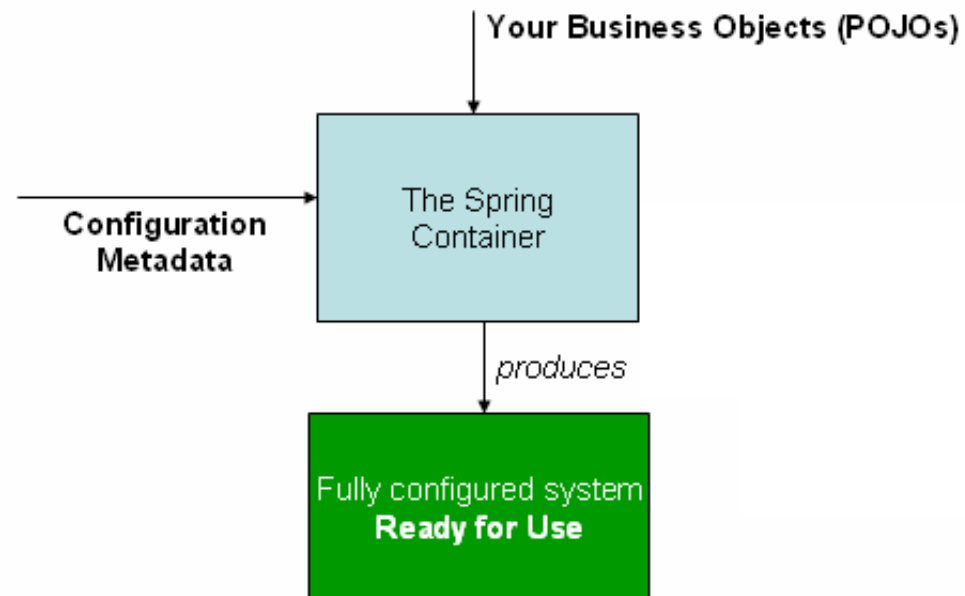
IoC container

- The Spring container (IoC Container) is the core of the Spring Framework.
- The container will create the objects, wire them together, configure them, and manage their complete lifecycle from creation till destruction



IoC container

- The container gets its instructions on what objects to instantiate, configure, and assemble by reading configuration metadata provided.
- The configuration metadata can be represented either by:
 - XML
 - Java annotations
 - Java code.



What is a Bean?

- Typical java bean with a unique id
- In spring there are basically two types
 - Singleton
 - One instance of the bean created and referenced each time it is requested
 - Prototype (non-singleton)
 - New bean created each time
 - Same as **new** ClassName()
- A Spring IoC container manages one or more beans
- Beans are normally created by Spring as late as possible



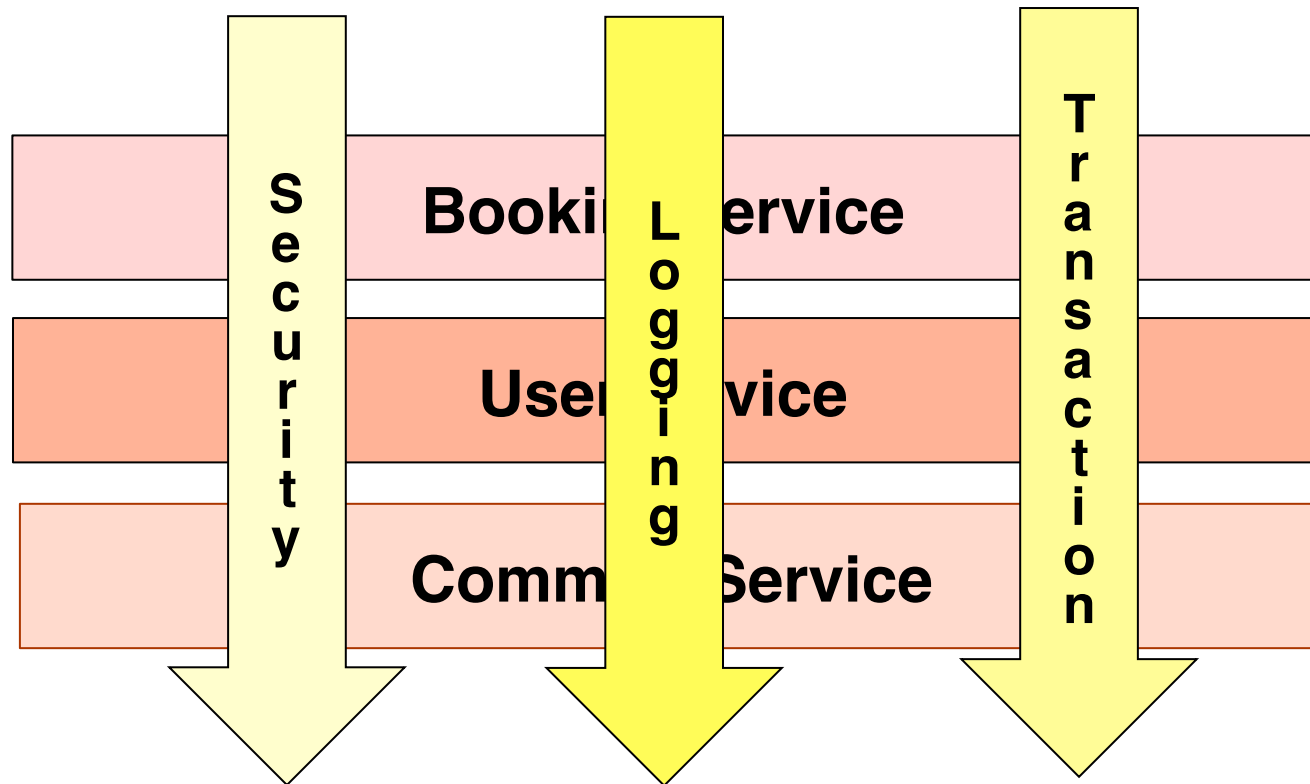
Aspect Oriented Programming (AOP)

- AOP entails breaking down program logic into distinct parts called concerns.
- The functions that span multiple points of an application are called cross-cutting concerns and these cross-cutting concerns are conceptually separate from the application's business logic.
- AOP is like triggers in programming languages such as Perl, .NET, Java and others.
- Examples of cross-cutting concerns:
 - Logging
 - Security
 - Transaction
 - Caching



Aspect Oriented Programming (AOP)

■ Cross-cutting



Aspect Oriented Programming (AOP)

```
package com.mkyong.aspect;

import org.aspectj.lang.JoinPoint;
import org.aspectj.lang.annotation.Aspect;
import org.aspectj.lang.annotation.Before;

@Aspect
public class LoggingAspect {

    @Before("execution(* com.mkyong.customer.bo.CustomerBo.addCustomer(..))")
    public void logBefore(JoinPoint joinPoint) {

        System.out.println("logBefore() is running!");
        System.out.println("hijacked : " + joinPoint.getSignature().getName());
        System.out.println("*****");

    }

}
```

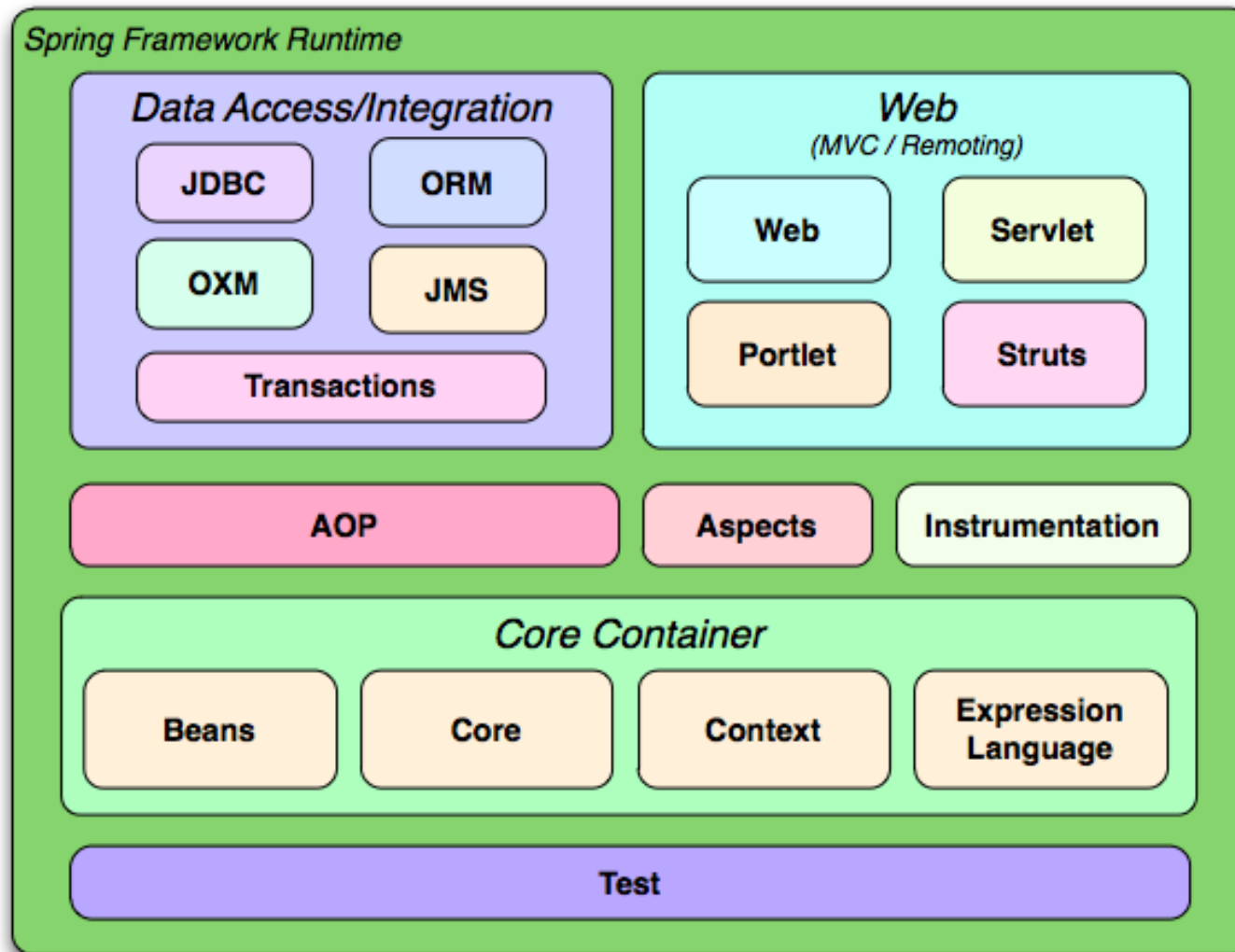
Pointcut (AspectJ Pointcut Expression Language)

Join Point

Advice



Spring Modules



- The Spring Framework consists of features organized into about 20 modules.
- These modules are grouped into Core Container, Data Access/Integration, Web, AOP (Aspect Oriented Programming), Instrumentation, and Test.

